D4 - Stockpile Block Model and Model Outputs

Pit I HG stockpile - current state after the 2026-03-29 survey (provisional through 2026-03-31)

1. Model files

File	Format	Use
`outputs/blockmodel/stockpile_block_model.csv`	CSV, one row per block containing material (2,699 blocks)	Excel, Vulcan, Datamine, Leapfrog, Deswik (import with centroid + size)
`outputs/blockmodel/stockpile_block_model.vtk`	VTK structured points, cell data	ParaView 3-D viewing (*Threshold* `tonnes > 0`)
`outputs/blockmodel/block_model_definition.json`	JSON	Origin, block size, dimensions, units, CRS note
`outputs/movement/*.asc`	ESRI ASCII grids (1 m)	Floor plane, pre-record surface, final model surface, latest surveyed surface (GIS / CAD)
`outputs/movement/load_destinations.csv`	CSV	Where each truck load was placed (destination centroid, push distance, spread)
`outputs/blockmodel/summary_*.csv`	CSV	Inventory by material source (**exact** class split), by dominant information class (whole blocks), confidence class, 4 m bench; grade-tonnage

Block definition. Origin (lower-left-bottom corner) E 511,290, N 7,122,280, Z 4,474.0. Blocks 5 × 5 × 2 m; 27 × 26 × 10 blocks. Partial blocks are exported with `fill_frac` and their true material volume. Coordinates are in the mine grid (see D2, A-01).

Units. Tonnes are as reported by the FMS (moisture basis not stated, A-26). Block volumes use the in-pile volume T × f_v / Dens with the calibrated volume factor f_v = 1.472. In the base case the in-pile density is 1.40 t/m³ (A-06; D2 Section 2.5). Cu is in %, Au and Ag in g/t (as supplied, see QA-F04). Contained Cu is in tonnes, Au and Ag in kg (1 koz = 31.1035 kg).

1.1 Block fields

Field	Meaning	Info type
`ix, iy, iz, xc, yc, zc, dx, dy, dz`	Block index, centroid, size	-
`volume_m3`, `fill_frac`	Material volume in block; fraction of block filled	Measured geometry + inferred where unsurveyed

Field	Meaning	Info type
`tonnes`, `density`	Total tonnes; derived in-pile density (tonnes ÷ volume)	Measured tonnes, calibrated volume (FMS); assumed by class otherwise
`cu_pct, au_gpt, ag_gpt`	Best-estimate grade (all classes)	Mixed: see class fractions
`cu_t, au_kg, ag_kg`	Contained metal	Mixed
`t_fms, t_unattributed, t_legacy, t_provisional`	Tonnes by information class (`t_legacy` = pre-record core)	-
`frac_fms, frac_unattributed, frac_legacy`	Class fractions	-
`cu_pct_fms, au_gpt_fms`	Grade of the FMS-tracked part only	Measured inputs, model-generated placement
`cu_sd_within, au_sd_within`	Grade heterogeneity among the loads mixed into the block	Model-generated
`n_eff_loads, n_loads_ge1pct, top_loads`	Effective number of loads; loads ≥ 1% of block; top three cycle IDs with share	Model-generated provenance
`mean_place_date`	Tonnage-weighted placement time of FMS material	Measured + model
`top_surveyed_frac`	Fraction of block footprint whose top surface was surveyed	Measured
`cu_p10/p50/p90, au_p10/p50/p90, tonnes_p10/p90, cu_cv, au_cv, tonnes_cv`	Uncertainty (Monte Carlo, 60 realisations; prior load-scale range for blocks < 50% FMS-tracked, see `grade_range_basis`)	Model-generated
`info_class`	Dominant class: FMS-reconciled / Unattributed-imputed / Legacy-assumed / Provisional	-
`confidence`	High / Medium / Low / Uninformed (rules in D3, Section 5.1)	-

2. Inventory

2.1 By information class (base case)

Inventory by information class (FMS grades/metal shown for the whole FMS class)

class	tonnes	share	Cu %	Au g/t	Cu (t)	Au (koz)	basis of grade
FMS-tracked, survey-reconciled	44,422	32.1%	8.42	8.00	3,760	11.5	measured inputs; location model-generated
FMS-tracked, provisional (after last survey)	228	0.2%					measured inputs; location from forward model

class	tonnes	share	Cu %	Au g/t	Cu (t)	Au (koz)	basis of grade
Unattributed (surveyed volume, no FMS record)	3,901	2.8%	10.22	8.82	399	1.1	imputed (same/adjacent intervals)
Record-period total	48,551	35.1%	8.57	8.06	4,159	12.6	
Pre-record core (legacy)	89,795	64.9%	8.42	8.00	7,562	23.1	assumed (FMS mean); geometry inferred
Whole pile modelled	138,346	100.0%	8.47	8.02	11,722	35.7	

> **Read the record-period total first.** It is the material placed during the FMS record (15-31 March) and is tied to surveyed fill. The pre-record core has inferred geometry and an assumed grade. The whole-pile total includes it.
>
> **Not included in any range in this table:** the interpretation that the 47% volume excess is loads missing from the FMS. In that case record-period tonnes would be 71,338 t (scenario X2, D6 Section 10.2).
>
> **Two class tables.** This table is the *exact* split: every block's tonnes are divided among the classes. `summary_by_dominant_info_class.csv` instead counts each whole block under its dominant class, for maps and reclaim planning, so its class totals differ even though the grand total is the same.
>
> **Identical grades across classes** follow from the assumptions: legacy grade = FMS mean (A-22), and unattributed grade is imputed from the same loads (A-21).
>
> **Grade units are as supplied and unconfirmed (QA-F04).**

> **Reading this table.** Only the first two rows rest on measured grade data. The unattributed row has a measured volume but an imputed grade. The pre-record core's grade is assumed, and its volume depends on assumptions A-13 and A-14.

2.2 By confidence class

Table 8.1 - Inventory by confidence class

confidence	blocks	tonnes	cu_pct	au_gpt	cu_t	au_koz
High	667	35,591	8.22	7.89	2,925	9.0
Medium	270	11,229	9.31	8.38	1,046	3.0
Low	122	6,574	8.97	8.30	590	1.8
Uninformed	1,640	84,951	8.43	8.01	7,161	21.9

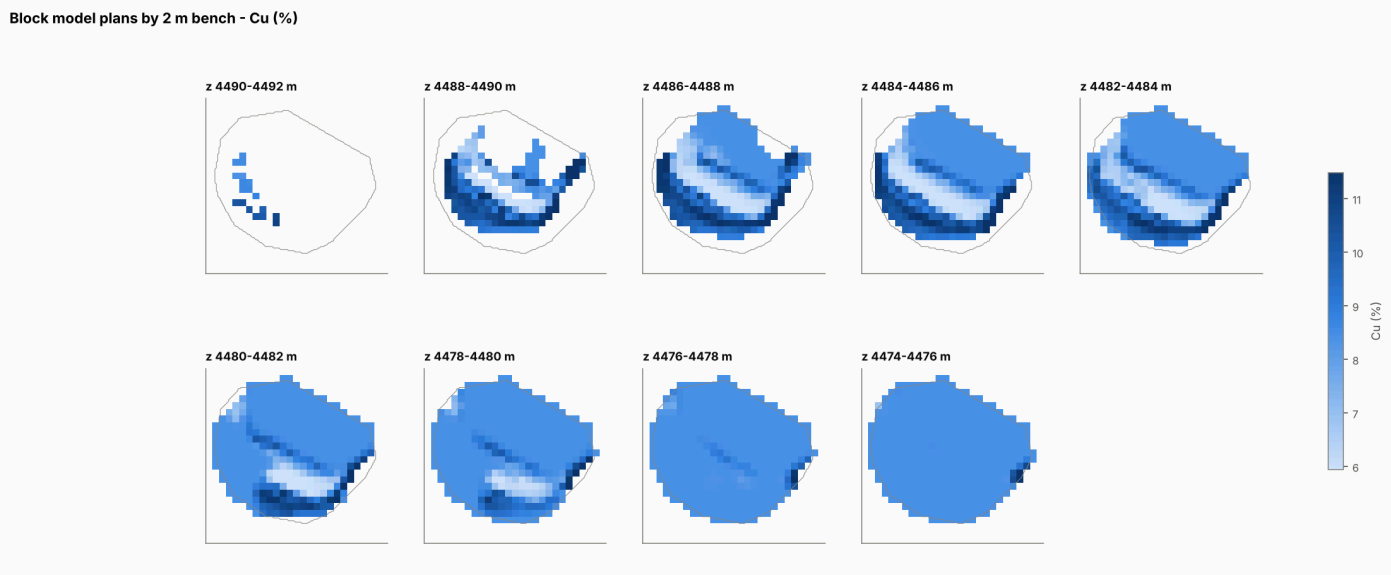
2.3 By 4 m reclaim bench

Table 8.2 - Inventory by 4 m reclaim bench (all classes; see D4 for class split)

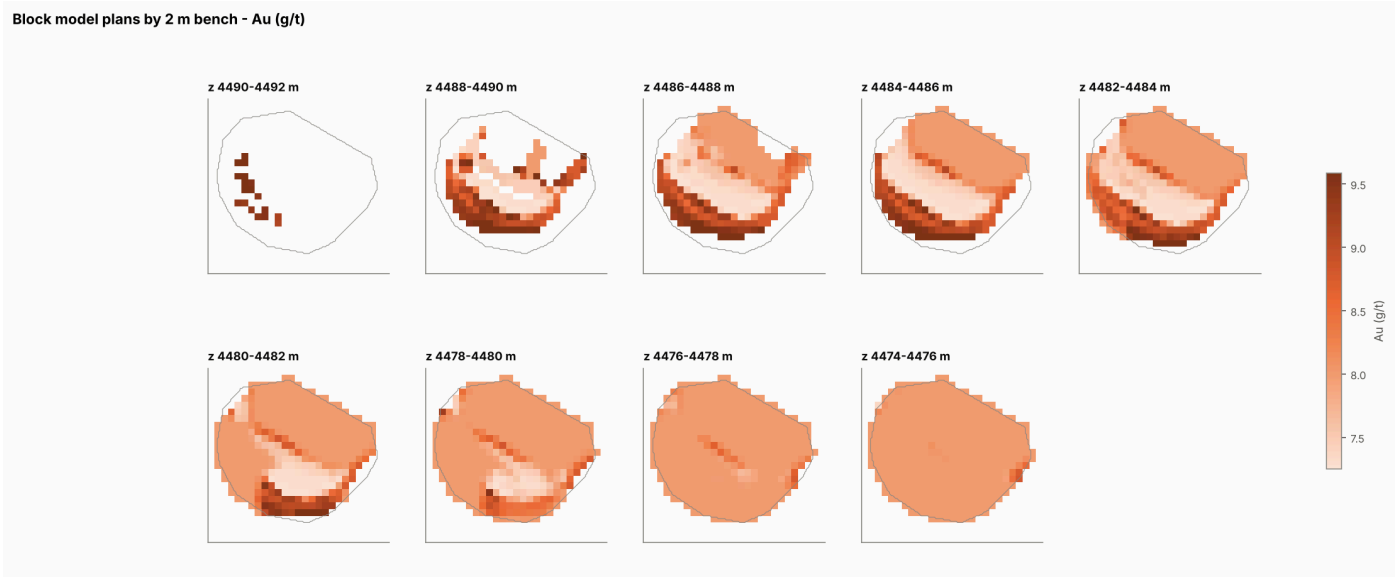
bench	blocks	tonnes	cu_pct	au_gpt	cu_t	au_koz
4490-4494	18	50	8.72	9.96	4	0.0
4486-4490	460	14,305	8.75	8.15	1,251	3.7
4482-4486	671	40,591	8.45	8.02	3,429	10.5
4478-4482	757	46,928	8.43	8.00	3,957	12.1
4474-4478	793	36,472	8.44	8.00	3,080	9.4

Source: `outputs/blockmodel/summary_by_bench_4m.csv` (model run 20261008T173954).

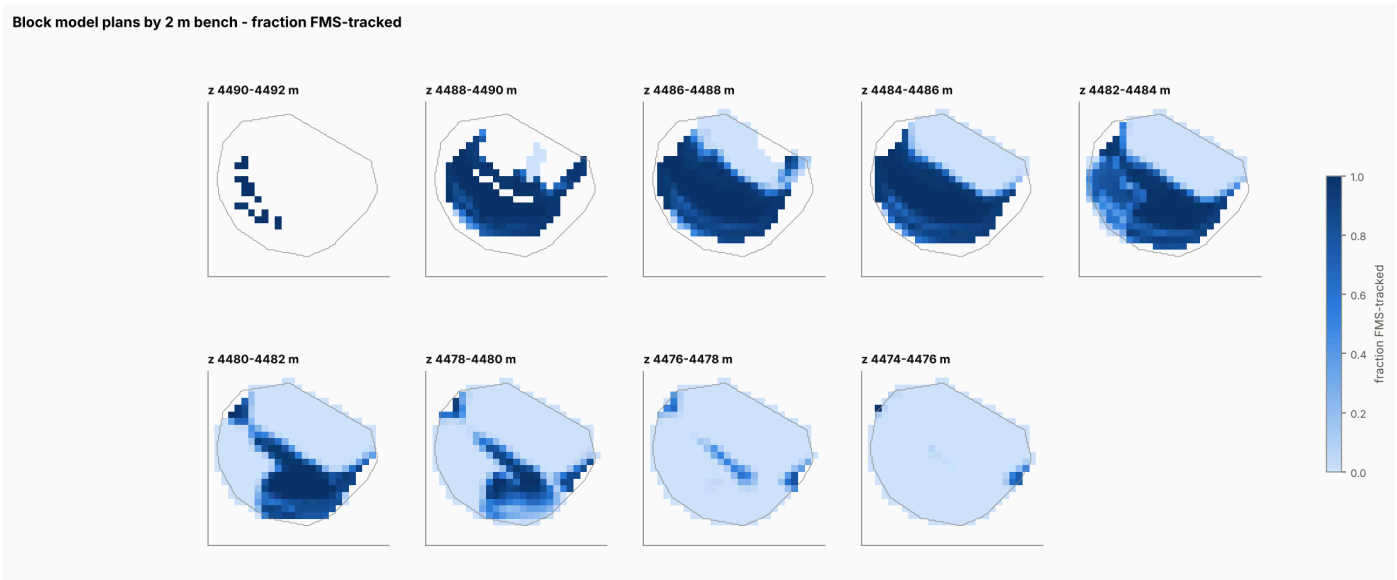
3. Visualisations



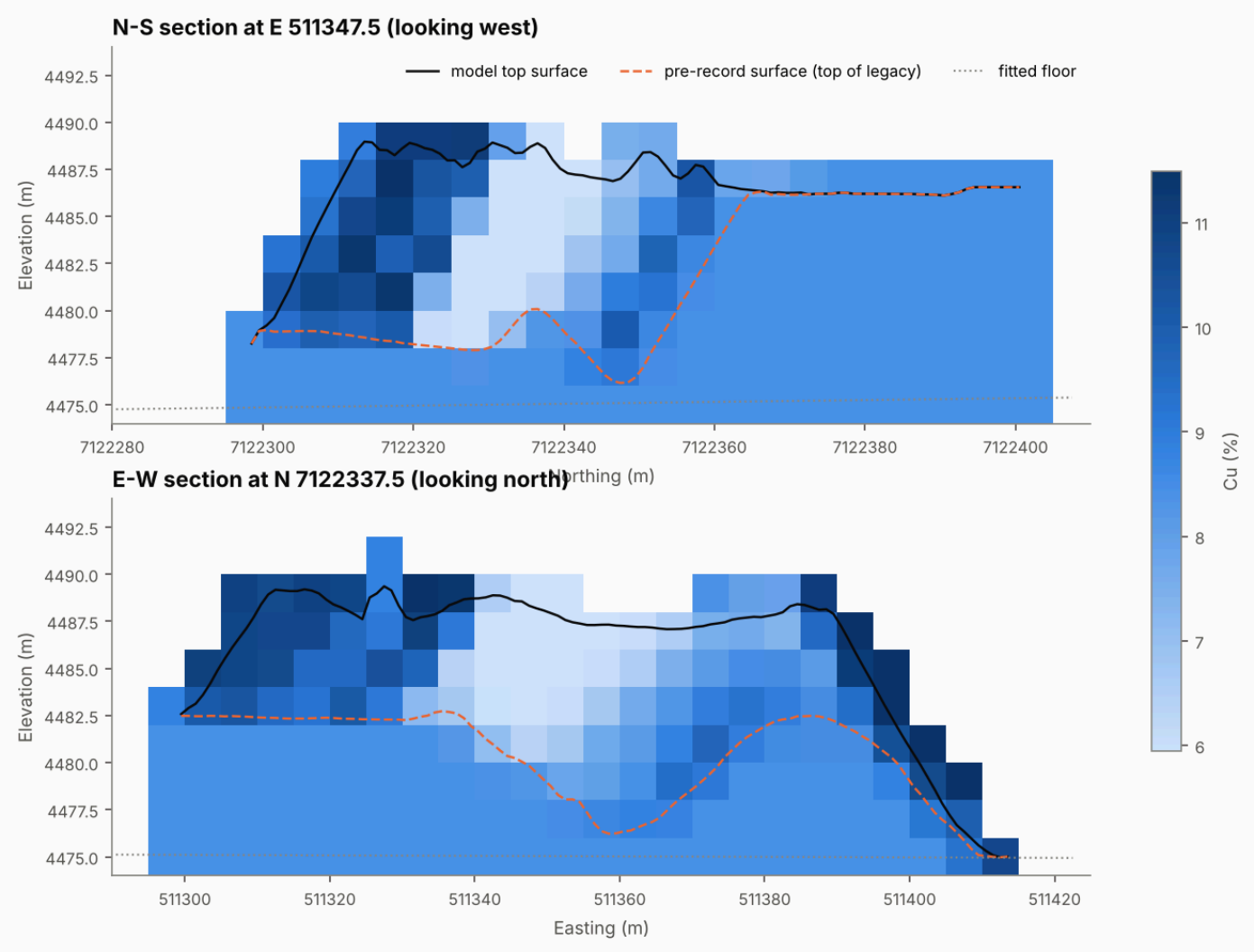
Cu grade by 2 m bench



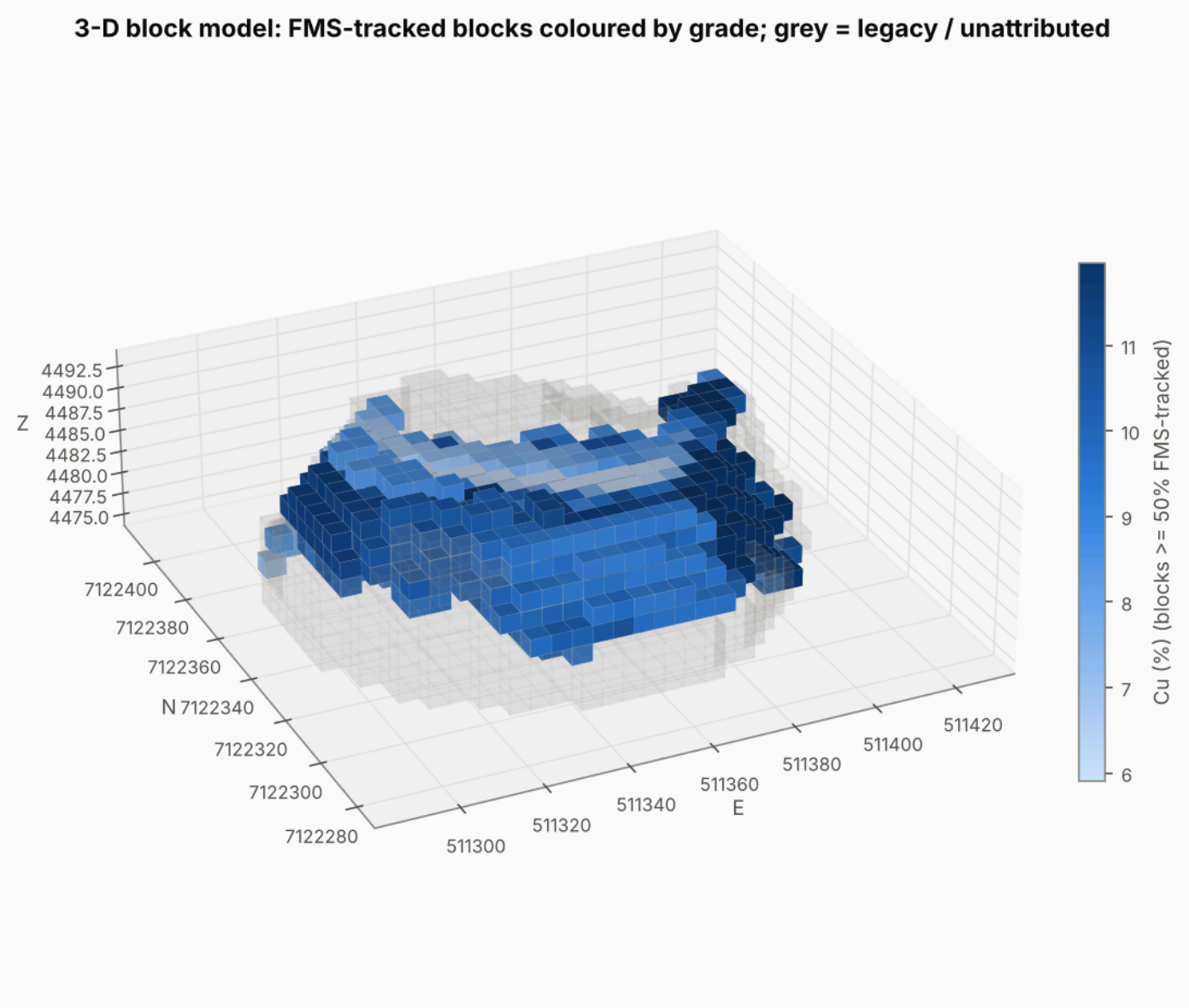
Au grade by 2 m bench



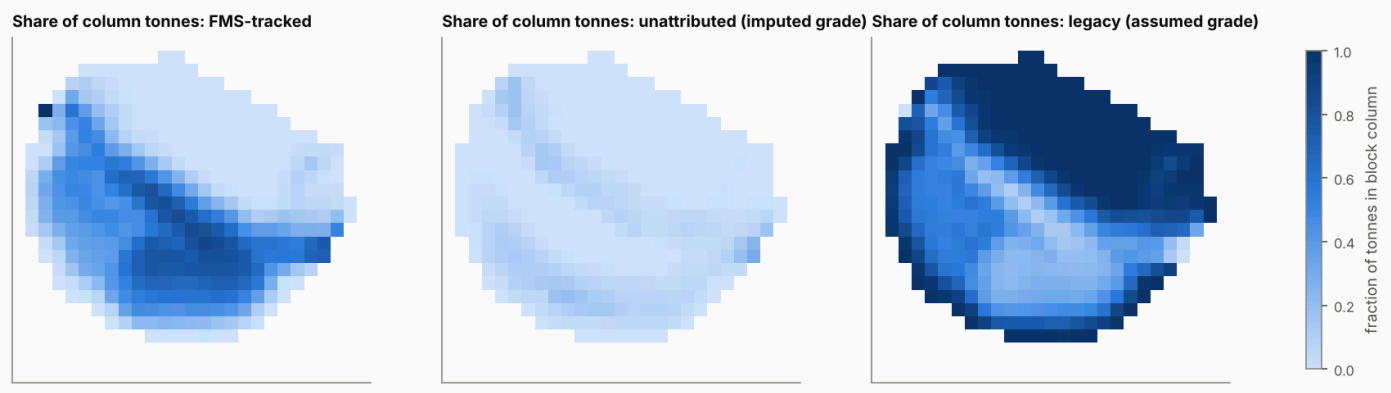
Fraction of block tonnes that is FMS-tracked



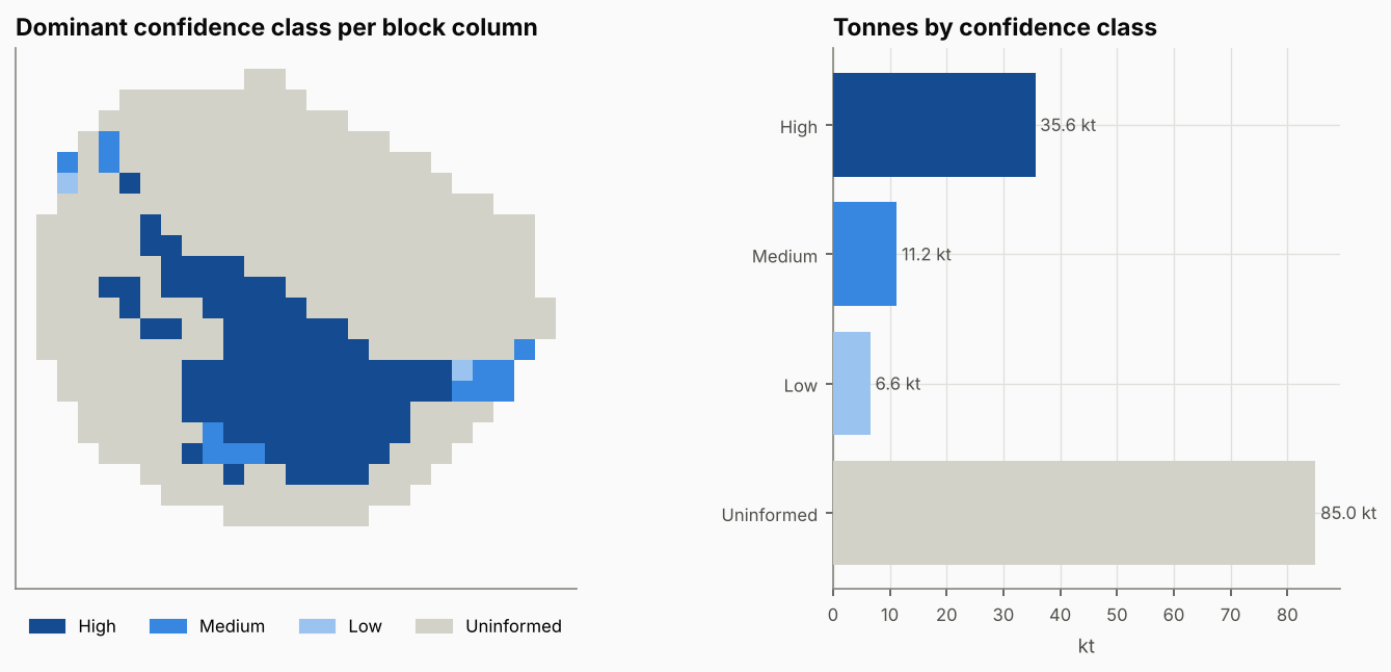
Sections through the tonnage-weighted centre of FMS material



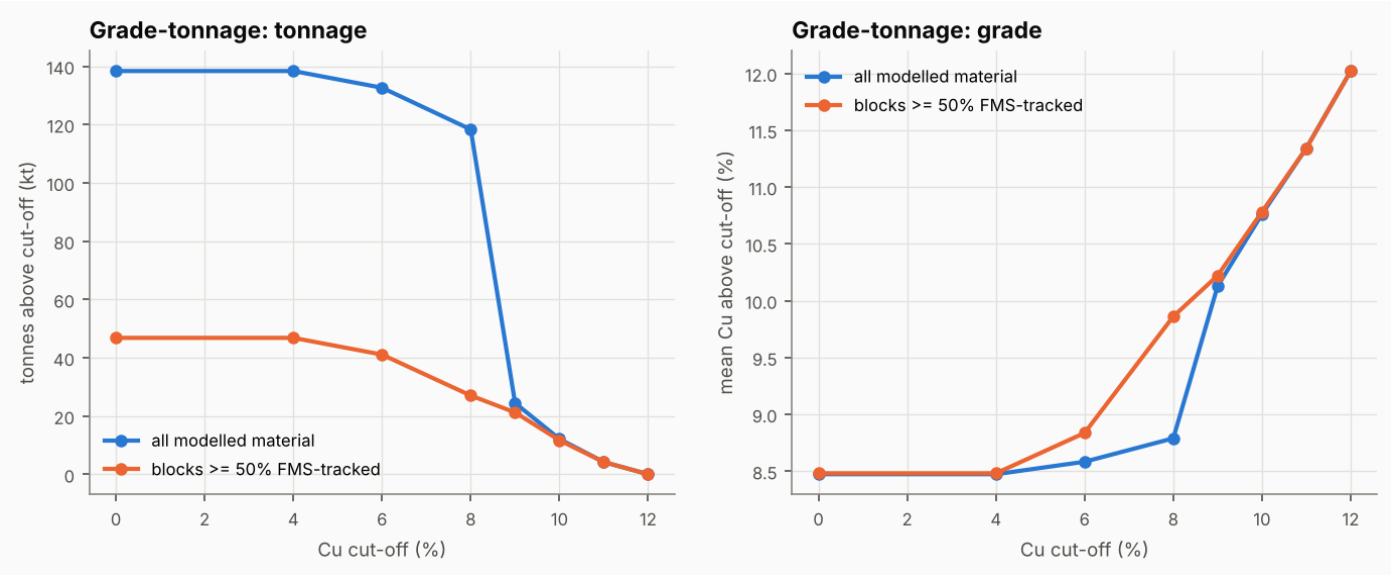
3-D view: FMS-tracked blocks coloured by Cu



Share of column tonnes by information class



Confidence classes



Grade-tonnage curves



Block uncertainty (Monte Carlo)

4. How to use the model

- **Reclaim sequencing / blending.** Filter `confidence ∈ {High, Medium}` for material whose grade rests on FMS data. Use `cu_p10` and `cu_p90` to test blends against plant constraints. Treat `Legacy-assumed` blocks as unknown-grade material until sampled.
- **Plant-feed / metal forecasting.** Use the bench summaries with their P10-P90 ranges. Report the record-period inventory first and separately from the pre-record core (D6, Section 13). Quote the scenario X2 next to any tonnage range.
- **Reconciliation.** `top_loads` and `load_destinations.csv` let mill or stacker grade measurements be compared with the loads predicted to be in a reclaimed block. This is how ε and the push parameters should be calibrated in operation.
- **Updating.** After each new survey run `python -m spbm update`. Provisional blocks will be re-reconciled.